

FREEDOM INTERNATIONAL SCHOOL
WORKSHEET- MCQ- SOLUTIONS
CLASS XI
PHYSICS
GRAVITATION

1. A body weighed 250 N on the surface. Assuming the earth to be a sphere of uniform mass density, how much would it weigh half way down to the centre of earth?

(a) 240 N (b) 210 N (c) 195 N (d) 125 N

Ans: (d) 125 N

$$\begin{aligned} W' &= mg' \\ &= m \times \frac{1}{2}g = \frac{1}{2}mg = \frac{1}{2}W \\ &= \frac{1}{2} \times 250 = 125 \text{ N} \end{aligned}$$

2. At what altitude (h) above the earth's surface would the acceleration due to gravity be one-fourth of its value at the earth's surface?

(a) $h = R$ (b) $h = 4R$ (c) $h = 2R$ (d) $h = 16R$

Ans: (a) $h = R$

$$g_h = g/4 = g \left(\frac{R}{R+h} \right)^2$$

$$\frac{R}{R+h} = \frac{1}{\sqrt{4}} = \frac{1}{2}$$

$$2R - R = h$$

$$R = h$$

3. Mass of moon is 1/81 times that of earth and its radius is 1/4 the earth's radius. If the escape velocity at the surface of earth is 11.2 km/s, then its value at the surface of moon is

(a) 0.14 km/s (b) 2.5 km/s (c) 0.5 km/s (d) 5 km/s

Ans: (b) 2.5 km/s

Here : Mass of moon $M_m = \frac{M_e}{81}$

Radius of moon $R_m = \frac{R_e}{4}$

Escape velocity on the surface of earth $V_{es(e)} = 11.2 \text{ km/s}$

The relation for escape velocity is

$$v = \sqrt{\frac{2GM}{R}} \propto \sqrt{\frac{M}{R}}$$

$$\text{Hence } \frac{v_{es(m)}}{v_{es(e)}} = \sqrt{\frac{M_m}{R_m} \times \frac{R_e}{M_e}}$$

$$= \sqrt{\frac{M_e}{81} \times \frac{4}{R_e} \times \frac{R_e}{M_e}} = \frac{2}{9}$$

$$v_{es(m)} = \frac{2}{9} \times 11.2 = 2.5 \text{ km/s}$$

4. A ball is dropped from a spacecraft revolving around the earth at a height of 120 km. What will happen to the ball?
- It will go very far in the space.
 - It will move with the same speed tangentially to the spacecraft.
 - It will fall down to the earth gradually.
 - It will continue to move with the same speed along the original orbit of spacecraft.

Ans: (d) It will continue to move with the same speed along the original orbit of spacecraft.

Since no external torque is applied therefore according to law of conservation of angular momentum the ball will continue to move with the same angular velocity along the original orbit of the spacecraft.

5. Two satellites of masses m_1 and m_2 (where m_1 is greater than m_2) are going around the earth in orbits of radii r_1 and r_2 ($r_2 > r_1$). Which statement about velocities is correct?
- $v_1 > v_2$
 - $v_1 < v_2$
 - $v_1 = v_2$
 - $v_1/r_1 > v_2/r_2$

Ans: (b) $v_1 < v_2$

For a satellite of mass m moving in an orbit of radius r ,

$$\frac{GM_e m}{r^2} = \frac{mv^2}{r}$$

$$\text{or } v = \sqrt{\frac{GM_e}{r}} \propto \sqrt{\frac{1}{r}} \quad \text{or } \frac{v_1}{v_2} = \sqrt{\frac{r_2}{r_1}}$$

Since $r_1 > r_2$, therefore $v_1 < v_2$.

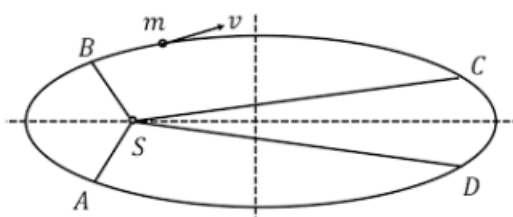
6. If the radius of earth's orbit is made one-fourth, then the duration of one year will become
- 8 times
 - $\frac{1}{4}$ times
 - 4 times
 - $\frac{1}{8}$ times

Ans: (d) $\frac{1}{8}$ times

$$\frac{T_1^2}{T_2^2} = \frac{R_1^3}{\left(\frac{R_1}{4}\right)^3} = \frac{64}{1} = 4^3$$

$$\therefore \frac{T_1}{T_2} = \left(4^{3/2}\right) = 8 \quad \therefore T_2 = \frac{T_1}{8}$$

7. The figure shows an elliptical orbit of a planet m about the sun S . The shaded area of SCD is twice the shaded area SAB . If t_1 is the time for the planet to move from C to D and t_2 is the time to move from A to B , then



- (a) $t_1 = 4t_2$ (b) $t_1 = 2t_2$ (c) $t_1 = t_2$ (d) $t_1 > t_2$

Ans: (b) $t_1 = 2t_2$

$$\begin{aligned} &\text{Law of areas,} \\ &\frac{\Delta A}{\Delta t} = \text{constant} \\ &\Delta A \propto \Delta t \\ &\frac{\Delta A_{\text{SCD}}}{\Delta A_{\text{SAB}}} = \frac{t_1}{t_2} \\ &\frac{2A}{A} = \frac{t_1}{t_2} \\ &\therefore t_1 = 2t_2 \end{aligned}$$

8. The kinetic energy needed to project a body of mass m from the earth's surface (radius R) to infinity is

- (a) $mgR/2$ (b) $2mgR$ (c) mgR (d) $mgR/4$

Ans: (c) mgR

The minimum kinetic energy required to project a body of mass m from earth's surface to infinity is known as escape energy. Therefore,

$$KE = \frac{GM_em}{R} = mgR$$

$$(\because gR = \frac{GM_e}{R})$$

9. The escape velocity for a body projected vertically upwards from the surface of earth is 11 km/s. If the body is projected at an angle 45° with the vertical, the escape velocity will be

- (a) $11/\sqrt{2}$ km/s (b) 11 km/s (c) $11\sqrt{2}$ km/s (d) 22 km/s

Ans: (b) 11 km/s

The escape velocity is independent of angle of projection, hence it will remain same i.e., 11 km/s.

10. A man is standing on an international space station, which is orbiting earth at an altitude 520 km with a constant speed 7.6 km/s. If the man's weight is 50 kg, his acceleration is

- (a) 7.6 km/s² (b) 7.6 m/s² (c) 8.4 m/s² (d) 10 m/s²

(c) 8.4 m/s²

A man of mass 50 kg is standing on an international space station.

Then, distance of station from centre of earth

$$r = R + h = 6400 + 520$$

$$[\because \text{radius of earth } R = 6400 \text{ km}]$$

$$= 6920 \text{ km}$$

$$\text{Now, centripetal acceleration } a = \frac{v^2}{r}$$

$$= \frac{(7.6 \times 10^3)^2}{6920 \times 10^3} = 8.4 \text{ ms}^{-2}$$

Directions: In the following questions, a statement of Assertion (A) is followed by a statement of Reason (R). Mark the correct choice as

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is NOT the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

11. **Assertion:** If a pendulum falls freely, then its time period is infinite.

Reason: Free falling body has acceleration equal to g .

Ans: b

12. **Assertion:** For the planets orbiting around the sun, angular speed, linear speed, K.E. changes with time, but angular momentum remains constant.

Reason: No torque is acting on the rotating planet. So, its angular momentum is constant.

Ans: a

13. **Assertion:** The value of acceleration due to gravity does not depend upon the mass of the body on which force is applied.

Reason: Acceleration due to gravity is a constant quantity.

Ans: c

14. **Assertion:** A body becomes massless at the centre of earth.

Reason: The acceleration due to gravity at a depth d is $g_d = g(1 - d/R)$

Ans: d

15. **Assertion:** The gravitational force on a particle inside a spherical shell is zero.

Reason: The shell shields other bodies outside it from exerting gravitational forces on a particle inside.

Ans: c